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Features & Functionality - Gym Management System

Functional Requirements

1. User Management

ID	Feature	Description	Priority
F1.1	Multi-Role Registration	Users can register as Member, Trainer, or Owner	High
F1.2	OAuth Integration	Login via Google, Facebook	High
F1.3	2FA Support	Email/SMS OTP verification	Medium
F1.4	Profile Management	Edit personal info, avatar, preferences	High
F1.5	Password Reset	Secure password recovery flow	High
F1.6	Account Status	Active, Suspended, Deleted states	Medium

2. Gym Administration

ID	Feature	Description	Priority
F2.1	Gym Creation	Owners can create and configure gyms	High
F2.2	Staff Management	Add/remove trainers, set roles	High
F2.3	Equipment Tracking	Inventory, maintenance scheduling	Medium
F2.4	Class Management	Group fitness scheduling	Medium
F2.5	Settings & Branding	Customize gym appearance	Low
F2.6	Invite System	Private gym invite codes	Medium

3. Membership Management

ID	Feature	Description	Priority
F3.1	Package Creation	Define membership packages with pricing	High

ID	Feature	Description	Priority
F3.2	Membership Purchase	Members can buy/renew memberships	High
F3.3	Approval Workflow	Owner approves membership requests	High
F3.4	Expiry Handling	Auto-expire, renewal reminders	High
F3.5	PT Session Credits	Include PT sessions in packages	Medium
F3.6	Membership Analytics	Track subscriptions, revenue	Medium

4. Personal Training

ID	Feature	Description	Priority
F4.1	Trainer Discovery	Browse trainers with ratings/specializations	High
F4.2	Trainer Assignment	Request and assign trainers	High
F4.3	Session Booking	Schedule PT sessions with availability	High
F4.4	Session Management	Complete, cancel, reschedule sessions	High
F4.5	Workout Plans	Trainers create customized plans	Medium
F4.6	Diet Plans	Nutritional guidance from trainers	Medium
F4.7	Session Ratings	Members rate completed sessions	Medium

5. Progress Tracking

ID	Feature	Description	Priority
F5.1	Body Measurements	Track weight, body fat, dimensions	High
F5.2	Progress Photos	Upload before/after images	Medium
F5.3	Workout Logging	Record daily workout activities	Medium
F5.4	Goal Setting	Define and track fitness goals	Medium
F5.5	Progress Charts	Visualize progress over time	Medium
F5.6	Achievement Badges	Gamification rewards	Low

6. Communication

ID	Feature	Description	Priority
F6.1	Real-time Chat	WebSocket-based messaging	High
F6.2	File Attachments	Share images, documents in chat	Medium
F6.3	Message Reactions	React to messages with emojis	Low
F6.4	Read Receipts	Track message delivery/read status	Low
F6.5	Notifications	In-app and push notifications	High
F6.6	Email Alerts	Configurable email notifications	Medium

7. Analytics & Reporting

ID	Feature	Description	Priority
F7.1	Owner Dashboard	Revenue, member stats, trends	High
F7.2	Trainer Reports	Session history, ratings, earnings	Medium

ID	Feature	Description	Priority
F7.3	Member Stats	Personal progress, attendance	Medium
F7.4	Financial Reports	Revenue breakdown, projections	Medium
F7.5	Export Reports	Download as PDF/CSV	Low

Non-Functional Requirements

1. Security

ID	Requirement	Implementation	Priority
NF1.1	Authentication	JWT tokens with expiry	Critical
NF1.2	Authorization	Role-Based Access Control (RBAC)	Critical
NF1.3	Password Security	BCrypt hashing (12 rounds)	Critical
NF1.4	Transport Security	HTTPS/TLS everywhere	Critical
NF1.5	Input Validation	Server-side validation	Critical
NF1.6	CORS Policy	Whitelist allowed origins	High
NF1.7	Rate Limiting	Prevent brute force attacks	High
NF1.8	SQL Injection Prevention	JPA Parameterized queries	Critical
NF1.9	XSS Prevention	Content sanitization	High

2. Performance

ID	Requirement	Target	Priority
NF2.1	API Response Time	< 200ms (95th percentile)	High
NF2.2	Page Load Time	< 3s initial, < 1s subsequent	High
NF2.3	Database Queries	Optimized with indexes	High
NF2.4	Caching	Spring Cache for frequent data	Medium
NF2.5	Connection Pooling	HikariCP (max 20 connections)	High
NF2.6	Lazy Loading	JPA relationships optimized	Medium

3. Scalability

ID	Requirement	Description	Priority
NF3.1	Multi-Tenancy	Isolated data per gym	High
NF3.2	Stateless Auth	JWT enables horizontal scaling	High
NF3.3	Database Scaling	Supports read replicas	Medium
NF3.4	WebSocket Scaling	Dedicated connection management	Medium
NF3.5	File Storage	External storage ready	Low

4. Reliability

ID	Requirement	Implementation	Priority
NF4.1	Data Integrity	Database transactions (ACID)	Critical
NF4.2	Soft Deletes	is_deleted flag, recoverable data	High
NF4.3	Optimistic Locking	@Version annotation	High
NF4.4	Error Handling	Global exception handler	High
NF4.5	Audit Logging	Track critical changes	Medium
NF4.6	Backup Strategy	Database backup support	High

5. Usability

ID	Requirement	Description	Priority
NF5.1	Responsive Design	Works on all screen sizes	High
NF5.2	Intuitive Navigation	Clear menu structure	High
NF5.3	Loading States	Skeleton loaders, spinners	Medium
NF5.4	Error Messages	User-friendly error display	High
NF5.5	Accessibility	WCAG 2.1 AA compliance	Medium
NF5.6	Internationalization	Multi-language support ready	Low

6. Maintainability

ID	Requirement	Description	Priority
NF6.1	Code Structure	Clean architecture layers	High
NF6.2	API Documentation	REST endpoints documented	Medium
NF6.3	Type Safety	TypeScript frontend, Java backend	High
NF6.4	Version Control	Git with branching strategy	High
NF6.5	Code Quality	Lombok for boilerplate reduction	Medium

Feature Matrix by Role

Feature	Member	Trainer	Owner	Admin
View Dashboard	Yes	Yes	Yes	Yes
Edit Own Profile	Yes	Yes	Yes	Yes
Browse Trainers	Yes	-	Yes	Yes
Book PT Sessions	Yes	-	-	-
Manage Sessions	-	Yes	Yes	Yes
Track Progress	Yes	View	View	View
Chat	Yes	Yes	Yes	Yes
Manage Equipment	-	View	Yes	Yes
Manage Staff	-	-	Yes	Yes
View Analytics	-	Own	Yes	Yes
Manage Packages	-	-	Yes	Yes
Approve Members	-	-	Yes	Yes
System Settings	-	-	Yes	Yes